MP204 – Electricity & Magnetism

Problem Set 1

Due in the assignment box by 5pm on Friday, 27 February 2015

1. The mass of the Earth is 5.97×10^{24} kg and that of the Moon is 7.35×10^{22} kg, and the distance between the two is 3.84×10^{8} m. Suppose we charged up both so that they each have an identical electric charge Q; find the value of Q (in coulombs) such that the resulting electric repulsion exactly equals the Earth-Moon gravitational attraction. (Use $G = 6.67 \times 10^{-11}\,\mathrm{N} \cdot \mathrm{m}^2 \cdot \mathrm{kg}^{-2}$ and $\epsilon_0 = 8.85 \times 10^{-12}\,\mathrm{C}^2 \cdot \mathrm{N}^{-1} \cdot \mathrm{m}^{-2}$.)

2. Consider the following charge configuration:

   (i) A charge $Q_1 = q$ placed at the origin $\vec{r}_1 = \vec{0}$;
   (ii) A charge $Q_2 = 2q$ placed at the point $\vec{r}_2 = a\hat{\imath} + a\hat{\jmath} + a\hat{k}$;
   (iii) A charge $Q_3 = -3q$ placed at the point $\vec{r}_3 = -a\hat{\jmath} + a\hat{k}$;

   where a is a constant. Using the Principle of Superposition, find expressions for the total electric field $\vec{E}(x, y, z)$ and total electric potential $\Phi(x, y, z)$ at the point $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$.

3. A square of side length a has three equal charges of size q placed at three of its vertices. We then place a fourth charge Q at the remaining vertex.

   (a) Determine the value of Q (given in terms of q) such that the total electric field at the square's centre is zero.

   (b) Determine the value of Q such that the total electric potential at the centre is zero.

   (c) Using the value of Q you obtained in (b), compute the electric field at the centre of the square.

4. (a) A system of charges has a potential

   $$\Phi(x, y, z) \quad = \quad 3x^2 y \sin(z).$$

   Find the electric field $\vec{E}(x, y, z)$.

   (b) A system of charges has electric field

   $$\vec{E}(x, y, z) \quad = \quad -x\hat{\imath} + y^2\hat{\jmath} + 3z\hat{k}.$$

   Find the potential $\Phi(x, y, z)$.